



# Beyond Improving Focus: Psychostimulant Medications as a Preventative Measure against Alzheimer's Disease in Adult ADHD Patients



## Abstract

This longitudinal study seeks to determine whether CNS stimulants provide a **protective effect against Alzheimer's disease (AD) in ADHD patients** by reducing oxidative stress and neuroaxonal damage, and subsequent dementia-related neurodegeneration.

It also aims to highlight the importance of **early, non-invasive AD screening** for a population at risk of late neurogenerative disease detection.

## Introduction

- **CNS stimulant medications (e.g. methylphenidate)** improve the symptoms of ADHD by increasing dopamine availability in the striatum and PFC (1, 2).
- Unmanaged ADHD increases the frequency of **negative lifestyle factors** associated with the development of AD (e.g. poor cardiovascular and metabolic health, alcohol abuse, psychosocial issues, depression, sleep problems) (3).
- It is hypothesised that **CNS stimulants may reduce chronic neuroinflammation** by improving ADHD symptom management and normalising brain iron levels (4).

Author: Holly M. Duncan  
Department: Psychology Department  
Institution: University of Essex Online  
Contact: hd26858@essex.ac.uk

## Research Question

Do central nervous system (CNS) stimulants provide a neuroprotective effect against the development of Alzheimer's disease in adults diagnosed with ADHD?

## Research Methods and Design

- **Sample size:** 44 (22 experimental, 22 control) (5).
- **Between subjects design:** Mixed ANOVA; experimental group (CNS stimulant-medicated); control group (unmedicated).
- **fMRI and QSM:** BOLD signal and brain tissue iron measurement.
- **Neuropsychological tests:** global cognition, working memory, episodic memory, verbal memory (6).
- **Tasks:** executive function, attention
- **Study timeline:** data collection at 0, 12, and 24 months.

Participants will undertake a series of **neuropsychological tests** designed to measure cognitive function while undergoing **fMRI and QSM imaging** procedures.

Cognitive function scores and brain iron load will be recorded during the tests in order to determine participants' **cognitive ability and level of neuroinflammation** respectively over time.

## ADHD: A Risk Factor for Dementia?

While not established as a direct risk factor, the symptoms of ADHD can mimic those of **Mild Cognitive Impairment (MCI)** in early AD (7):

- Working memory deficits
- Executive dysfunction
- Emotional dysregulation
- Anxiety
- Depression
- Sleep disturbances

These symptoms have the potential to 'mask' the prodromal stage of AD pathology in adults with ADHD. This may result in:

- **Delayed AD detection** and diagnosis
- Increased **emotional and psychological trauma** for patients and their families
- **Poorer clinical outcomes** due to delays in, or lack of, appropriate AD treatments

## Ethics

- The study will adhere to **BPS ethical guidelines** (8).
- Researcher collaboration with **qualified radiographers** is mandatory for data collection.
- **Encrypted data storage** is required to ensure confidentiality and anonymity.
- All participants have the **right to withdraw** from the study at any time.
- **Informed consent** obtained from all participants
- All relevant protocols reviewed as part of the **UoEO ethical approval process**.

## Study Limitations

- The influence of **additional medications** (e.g. SSRIs) and **non-drug therapies** (e.g. CBT)
- Participant use of **recreational drugs**
- Reliance on participants' consistent **self-administration** of CNS stimulants
- Participants' need to alter their **CNS stimulant dosage** during their course of treatment

## Take Home Message

Psychostimulant medications may offer ADHD patients protection against AD via reductions in contributing lifestyle factors and neuroaxonal damage.

## References

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